



RETROSPECTIVE RESEARCH

COMPARISON OF OUTCOMES IN LUMBAR STENOSIS CANAL PATIENTS WHO RECEIVED OPEN SURGERY WITH ENDOSCOPIC SURGERY

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INTRODUCTION



- Canal stenosis, a prevalent condition characterized by the narrowing of the spinal canal, has long been a significant challenge in the field of orthopedics.
- Lumbar spinal stenosis, the most common form, affects approximately 8-11% of individuals over the age of 50.
- Over the last decade, surgical techniques and instrumentation have been developed to reduce the invasiveness of surgery from classic open surgical decompression to full endoscopic decompression. Concerns about reducing the invasiveness of these procedures include epidural scarring and also reducing the likelihood of instability that will occur due to excessive resection of bone and soft tissue as well as muscle injury.
- Endoscopic surgery has emerged as a promising approach for the management of canal stenosis.



RESEARCH DESIGN

This retrospective study uses a descriptive and analytical design, gathering data retrospectively to compare outcomes in patients with lumbar canal stenosis who were treated with open surgery and endoscopic surgery at Dr. Soetomo General Hospital.

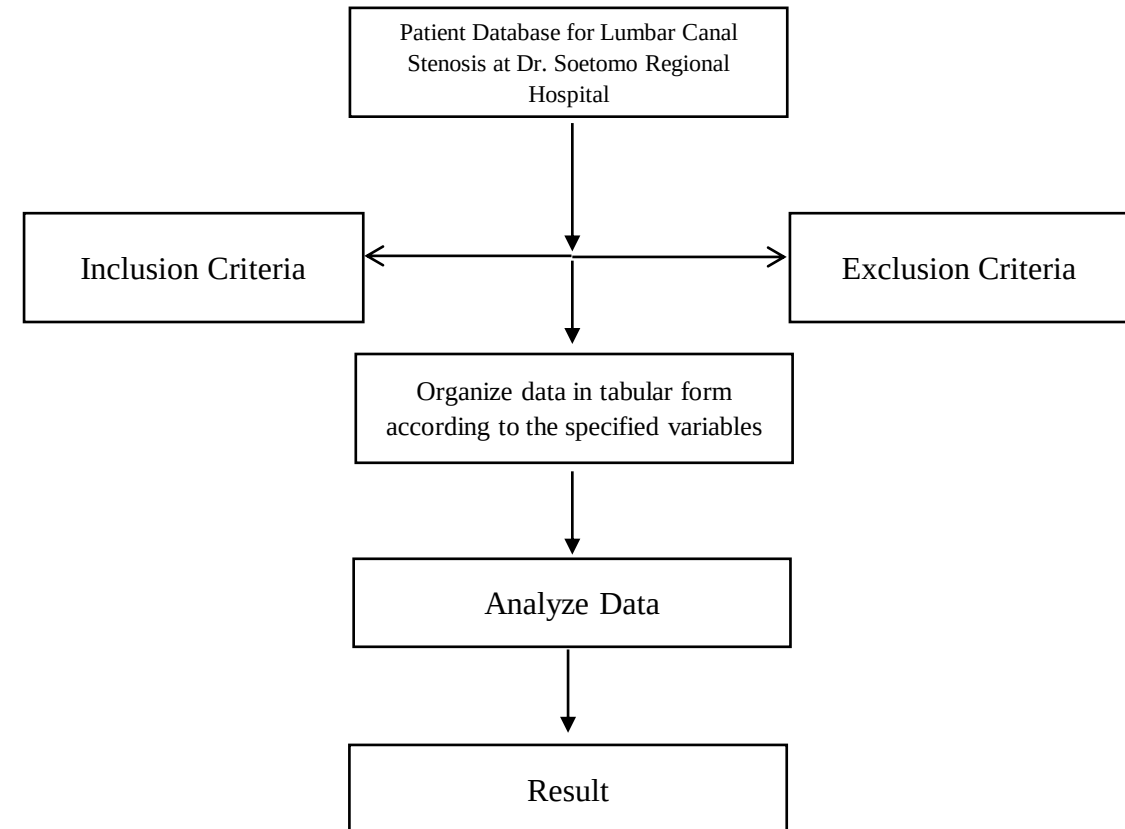
RESEARCH METHODOLOGY

- The researcher collects data on patients with lumbar canal stenosis who underwent open surgery and endoscopic surgery from the Medical Records Section of Dr. Soetomo Hospital Surabaya during the 2022-2023 period.
- The data is then processed using inclusion and exclusion criteria and assessed through subjective (Short Form Survey-12, Roland Morris Score, Oswestry Disability Index) and objective (Frankel Score) measures.

RESEARCH FLOW



- The retrospective study was conducted from February 2022 to April 2023, encompassing all stages from permitting to data collection.
- The study involved reviewing medical records obtained from the Medical Records Department of Dr. Soetomo General Hospital Surabaya for variables such as age, sex, education status, injury mechanism, type and level of injury, type of treatment, and patient outcomes.
- The sample size was based on data from the Spine Division of the Orthopaedics & Traumatology Department, with 32 samples available during the data collection period. Among the available samples, 26 data (81.2%) were deemed valid according to the predefined inclusion and exclusion criteria.
- Data analysis using the Chi-square test was conducted to assess the differences in proportions of categorical data for the Frankel scores.



RESULT



- Open surgery is used more often than endoscopic surgery for treating lumbar canal stenosis.
- Lumbar canal stenosis is most common in people aged 55-74. Patients who have endoscopic surgery tend to be older than those who have open surgery.
- Women are more likely to have lumbar canal stenosis, but the difference isn't significant.
- Most cases of lumbar canal stenosis occur in patients with a high school education.
- Patients with lumbar canal stenosis who have either open or endoscopic surgery mostly experience mild disability, according to the Short Form Survey 12, Roland Morris Score, and Oswestry Disability Index.



RESULT



- In terms of both physical and mental health, patients who have open surgery for lumbar canal stenosis generally experience mild disability.
- For physical health, patients who have endoscopic surgery for lumbar canal stenosis are equally likely to have mild disability or none at all. For mental health, most patients do not experience any disability.
- Patients with lumbar canal stenosis who undergo either open or endoscopic surgery mostly report mild disability related to back pain.
- Patients who have open surgery for lumbar canal stenosis are more likely to have a Frankel E neurological status compared to those who have endoscopic surgery.



RESULT

- There is a clear improvement in disability from before to after surgery, as shown by the SF-12, ODI, and Roland Morris Score questionnaires.
- Neurological status also improves from before to after surgery, according to the Frankel classification, for both open and endoscopic surgery groups.



CONCLUSION

- Open surgery remains the gold standard for lumbar canal stenosis, but research shows endoscopy can be equally effective.
- Endoscopy offers benefits like smaller incisions, less soft tissue damage, and quicker healing, but has drawbacks including limited indications and a narrower operative field, typically suited for single-level stenosis.



SUGGESTION

- Further research with a longer time period is needed to obtain a more comprehensive assesment of the outcomes of open surgery and endoscopic surgery in cases of lumbar canal stenosis at Dr. Soetomo General Hospital.
- It is also important to assess the need for improvements in standards and the digitization of the medical record system at Dr. Soetomo General Hospital, due to issues encountered during the research, such as incomplete medical record data and inadequate storage systems.





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